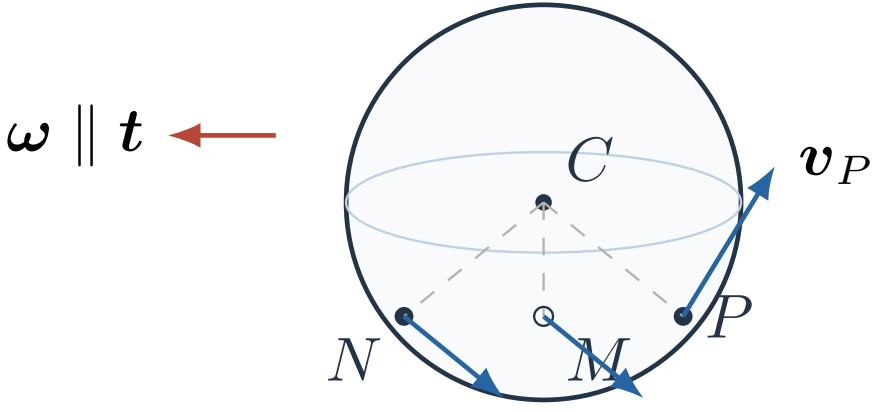
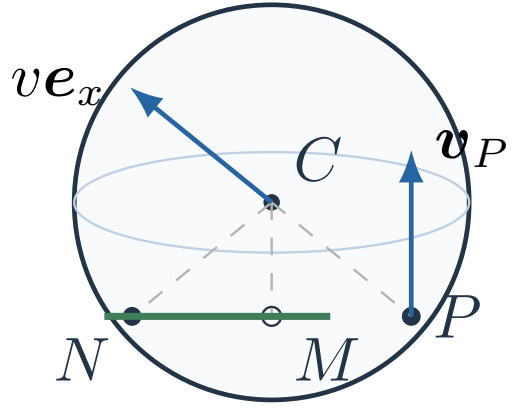
固定球心打滑



$\boldsymbol{v}_M, \boldsymbol{v}_N \parallel -\boldsymbol{e}_x$

$\boldsymbol{v}_C = 0$，摩擦与各点滑速反向

两壁纯滚动



$\boldsymbol{v}_M = \boldsymbol{v}_N = 0$

$|\boldsymbol{\omega}| = \sqrt{2}v/r$，$P$ 接触分离